

NANOTECHNOLOGY AND APPLICATIONS

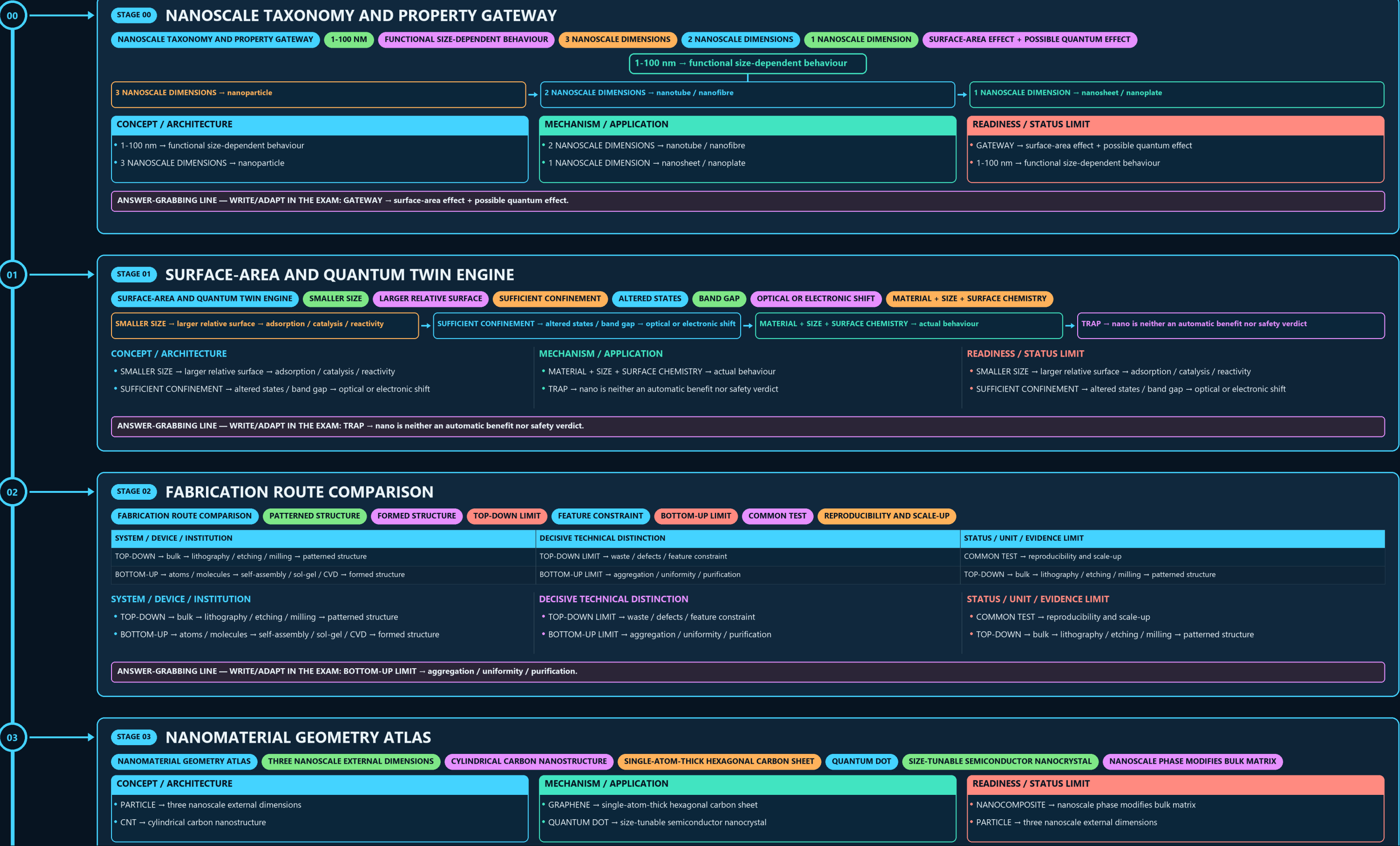
NANOSCALE TAXONOMY AND PROPERTY → SURFACE-AREA AND QUANTUM TWIN → FABRICATION ROUTE COMPARISON → NANOMATERIAL GEOMETRY ATLAS → CHARACTERIZATION DECISION TREE → NANOMEDICINE BARRIER RAIL

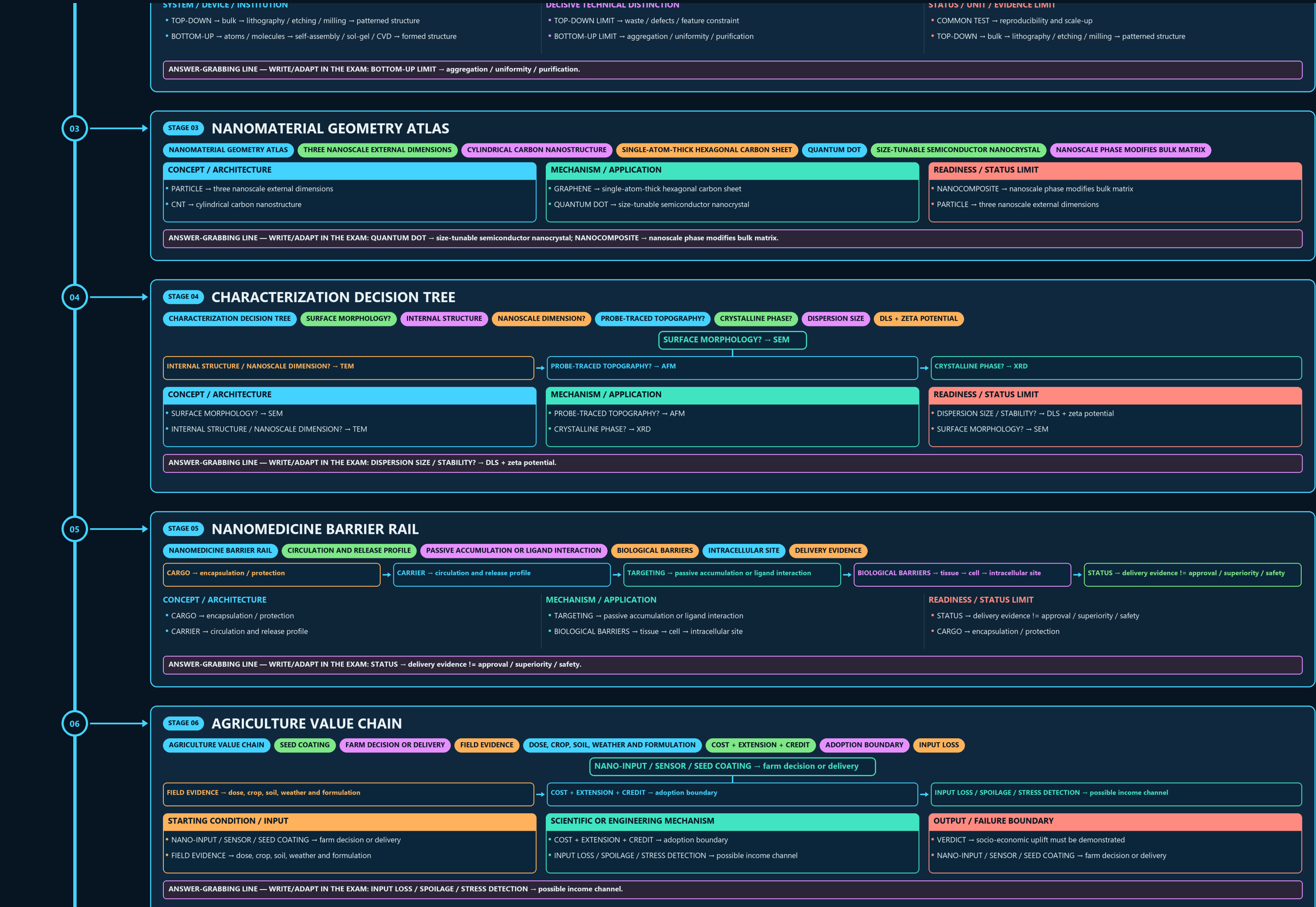
Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles





06

STAGE 06 AGRICULTURE VALUE CHAIN

AGRICULTURE VALUE CHAIN

SEED COATING

FARM DECISION OR DELIVERY

FIELD EVIDENCE

DOSE, CROP, SOIL, WEATHER AND FORMULATION

COST + EXTENSION + CREDIT

ADOPTION BOUNDARY

INPUT LOSS

NANO-INPUT / SENSOR / SEED COATING → farm decision or delivery

FIELD EVIDENCE → dose, crop, soil, weather and formulation

COST + EXTENSION + CREDIT → adoption boundary

INPUT LOSS / SPOILAGE / STRESS DETECTION → possible income channel

STARTING CONDITION / INPUT

- NANO-INPUT / SENSOR / SEED COATING → farm decision or delivery
- FIELD EVIDENCE → dose, crop, soil, weather and formulation

SCIENTIFIC OR ENGINEERING MECHANISM

- COST + EXTENSION + CREDIT → adoption boundary
- INPUT LOSS / SPOILAGE / STRESS DETECTION → possible income channel

OUTPUT / FAILURE BOUNDARY

- VERDICT → socio-economic uplift must be demonstrated
- NANO-INPUT / SENSOR / SEED COATING → farm decision or delivery

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INPUT LOSS / SPOILAGE / STRESS DETECTION → possible income channel.

07

STAGE 07 WATER AND ENVIRONMENT TREATMENT LOOP

WATER AND ENVIRONMENT TREATMENT LOOP

SYSTEM TEST

SELECTIVITY + FOULING + ENERGY

REGENERATION OR SPENT-MATERIAL CAPTURE

LIFECYCLE CHECK

SECONDARY NANOPARTICLE RELEASE

LAB REMOVAL

FIELD DEPLOYMENT

CONCEPT / ARCHITECTURE

- CONTAMINANT → detection / adsorption / membrane / photocatalysis
- SYSTEM TEST → selectivity + fouling + energy

MECHANISM / APPLICATION

- RECOVERY → regeneration or spent-material capture
- LIFECYCLE CHECK → secondary nanoparticle release

READINESS / STATUS LIMIT

- BOUNDARY → lab removal != field deployment
- CONTAMINANT → detection / adsorption / membrane / photocatalysis

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SYSTEM TEST → selectivity + fouling + energy.

08

STAGE 08 ELECTRONICS ENERGY MATERIALS MATRIX

ELECTRONICS ENERGY MATERIALS MATRIX

CONDUCTIVE AND STRUCTURAL RESEARCH

QUANTUM DOT

SENSING RESEARCH

COATINGS AND LIGHTWEIGHT-STRENGTH MODIFICATION

NANOSTRUCTURED ELECTRODE

STORAGE AND CONVERSION INTERFACES

CERTIFIED MANUFACTURABLE PRODUCT

SYSTEM / DEVICE / INSTITUTION

GRAPHENE / CNT → conductive and structural research

QUANTUM DOT → optical / display / sensing research

DECISIVE TECHNICAL DISTINCTION

NANOCOMPOSITE → coatings and lightweight-strength modification

NANOSTRUCTURED ELECTRODE → storage and conversion interfaces

STATUS / UNIT / EVIDENCE LIMIT

BOUNDARY → property != certified manufacturable product

GRAPHENE / CNT → conductive and structural research

SYSTEM / DEVICE / INSTITUTION

- GRAPHENE / CNT → conductive and structural research
- QUANTUM DOT → optical / display / sensing research

DECISIVE TECHNICAL DISTINCTION

- NANOCOMPOSITE → coatings and lightweight-strength modification
- NANOSTRUCTURED ELECTRODE → storage and conversion interfaces

STATUS / UNIT / EVIDENCE LIMIT

- BOUNDARY → property != certified manufacturable product
- GRAPHENE / CNT → conductive and structural research

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUANTUM DOT → optical / display / sensing research.

09

STAGE 09 INDIAN NANO INSTITUTION MAP

INDIAN NANO INSTITUTION MAP

DST NPNST

EARLIER NANO MISSION

PUBLIC R&D ARCHITECTURE

INST MOHALI

DST AUTONOMOUS NANOSCIENCE INSTITUTION

CSIR LABS

CROSS-SECTOR TRANSLATIONAL CAPABILITY ROUTE

DST NPNST / EARLIER NANO MISSION → public R&D architecture

INST MOHALI → DST autonomous nanoscience institution

CSIR LABS → cross-sector translational capability route

MeitY-LINKED NANO-ELECTRONICS → device / semiconductor bridge

CONCEPT / ARCHITECTURE

- DST NPNST / EARLIER NANO MISSION → public R&D architecture
- INST MOHALI → DST autonomous nanoscience institution

MECHANISM / APPLICATION

- CSIR LABS → cross-sector translational capability route
- MeitY-LINKED NANO-ELECTRONICS → device / semiconductor bridge

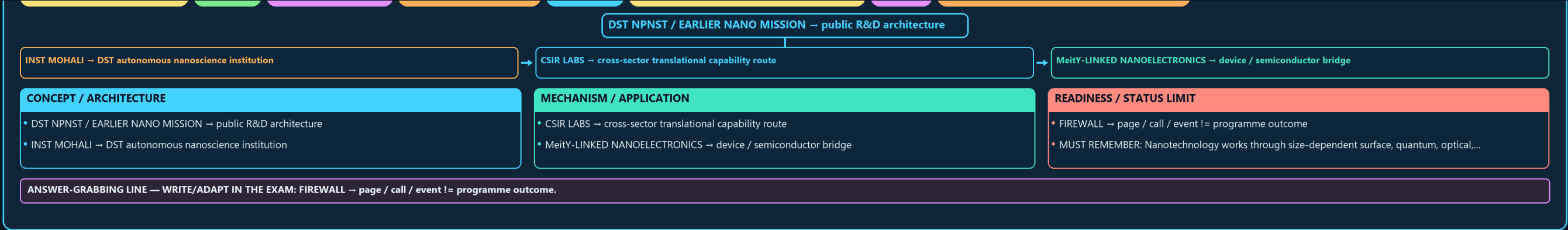
READINESS / STATUS LIMIT

- FIREWALL → page / call / event != programme outcome
- MUST REMEMBER: Nanotechnology works through size-dependent surface, quantum, optical,...

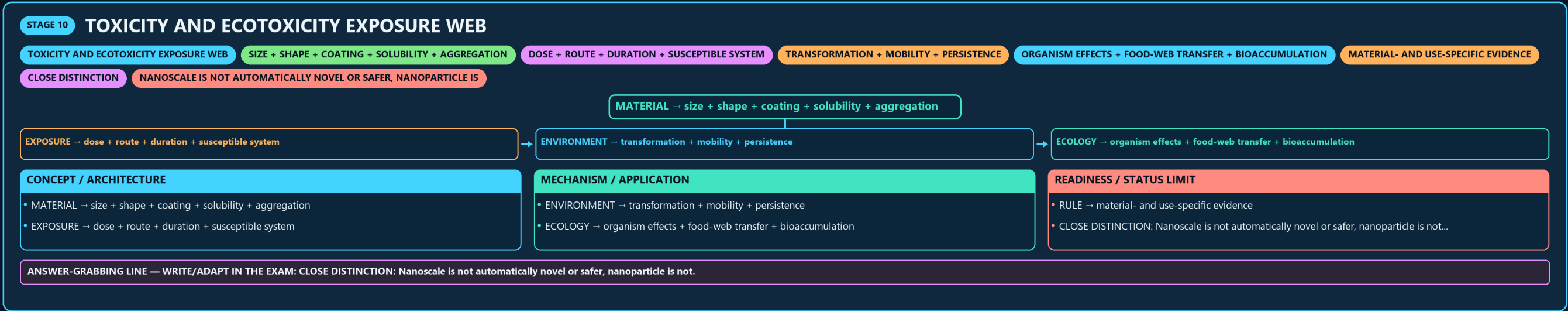
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FIREWALL → page / call / event != programme outcome.

10

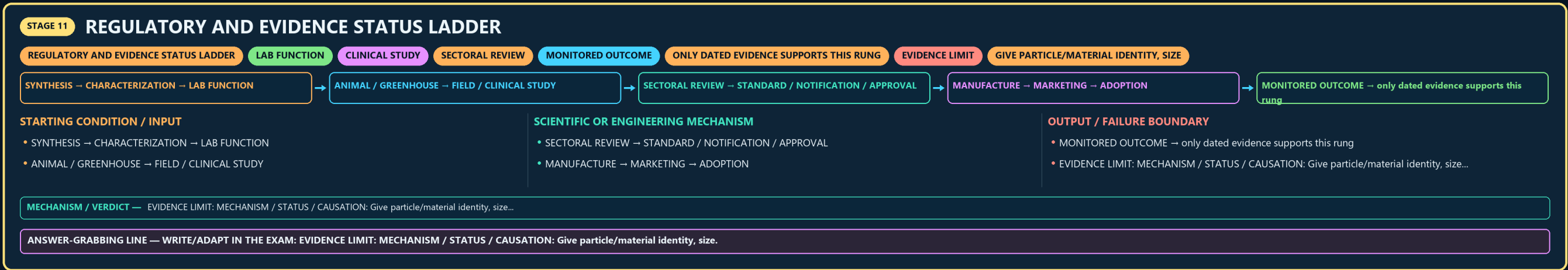
STAGE 10 TOXICITY AND ECOTOXICITY EXPOSURE WEB



10



11



E

